SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning / User Story Workshop

Course Code	Course Title	Assessment
CSA10 / CSA1005	Software Engineering	Agile Sprint Planning – User Story Workshop
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Date: 27-10-2026	Problem Statement: 7	Smart School Bus Tracking and Student Safety System

Annexure A – Agile Sprint Planning and User Story Workshop

1. Problem Statement and Project Vision

Schools need a reliable way to track buses, verify student boarding and drop-off, and inform parents about delays or route changes. The proposed Smart School Bus Tracking and Student Safety System combines GPS tracking, route management, attendance events, parent notifications and incident reporting in a secure web and mobile platform.

Product vision: Provide safe, transparent and accountable school transportation through real-time visibility and timely alerts.

2. Agile Team and Stakeholders

Scrum role / stakeholder	Responsibilities
Product Owner – Transport Coordinator	Prioritizes safety requirements, accepts stories, clarifies route and alert policies.
Scrum Master	Facilitates sprint planning, daily Scrum, reviews and retrospectives; removes blockers.
Development Team	Builds parent app, dashboard, GPS API, database, attendance and alert services; tests increments.
Parents, Drivers, Attendants, School Admin	Provide user feedback, validate usability, safety workflows and operational accuracy.

3. User Story Workshop

User stories follow: **As a [role], I want [feature], so that [benefit].** Priority uses MoSCoW: Must, Should, Could.

ID	Role	User story: I want to…	So that…	Priority	Pts
US01	Parent	View live bus location and ETA	Know when to reach the stop	Must	8
US02	Attendant	Record boarding and drop-off by scanning ID	Confirm each child's travel status	Must	5
US03	Coordinator	Create routes and assign buses, drivers and students	Manage daily transport	Must	8

ID	Role	User story: I want to…	So that…	Priority	Pts
US04	Parent	Receive boarding, drop-off and delay alerts	Stay informed about safety	Must	5
US05	Administrator	Receive route deviation alerts	Respond quickly to unexpected detours	Must	8
US06	Driver	Start/end trips and report incidents	Keep trip records accurate	Should	5
US07	Administrator	Generate attendance and punctuality reports	Review transport performance	Should	5
US08	Parent	Use a simple multilingual interface	Access information easily	Could	3

4. Acceptance Criteria for Key Stories

US01 – Live bus tracking

• Given an active trip, when a parent opens the app, the assigned bus appears on a map.

• The displayed position refreshes when new GPS data arrives; the last-update time is visible.

• Only authorized guardians can view the bus for their assigned student.

US02 – Boarding / drop-off

• Given an assigned student, when an attendant scans the ID, the correct event is recorded with time, bus and stop.

• Duplicate scans are identified and the attendant receives a clear confirmation.

• The corresponding parent can view the latest verified event.

US05 – Route deviation alert

• Given an active route, when the bus remains outside the allowed route corridor for three readings, an alert is created.

• The coordinator sees the affected bus, timestamp and location.

• The alert is resolved or updated when the bus returns to its route.

5. Product Backlog and Prioritization

Priority	Backlog items	Reason
P1 – Critical	Login/roles, route assignments, GPS tracking, boarding/drop-off	Core student safety and access control.
P2 – High	Parent notifications, ETA, deviation and delay alerts	Immediate visibility and rapid response.
P3 – Medium	Incident management, reports, audit history	Accountability and operational improvement.
P4 – Future	Traffic-aware ETA, multilingual UI, advanced analytics	Additional value after core reliability is proven.

6. Sprint Planning

Sprint duration: 2 weeks. **Initial capacity:** 20 story points per sprint (planning assumption, reviewed using actual velocity). The Product Owner sets priorities; developers estimate and select feasible work.

Sprint	Sprint goal	Selected work	Expected increment
0	Prepare foundation	Requirements, architecture, database schema, CI and backlog	Reviewed design and ready development environment
1	Manage transport records	Role-based login, routes, buses, drivers, students (US03)	Working admin dashboard
2	Make bus location visible	GPS API, live map, ETA (US01)	One-bus live-tracking pilot
3	Verify student journeys	Boarding/drop-off (US02), parent alerts (US04)	Attendance and notifications demo
4	Respond to disruption	Deviation alerts (US05), trip/incident reporting (US06)	Operational safety alert workflow
5	Review and release	Reports (US07), testing, security, user acceptance	Pilot-ready tested release

7. Detailed Sprint 2 Task Breakdown

Task	Owner	Estimated effort	Deliverable
Define GPS payload and secure endpoint	Backend developer	2 days	Validated GPS ingestion API
Store location and trip history	Backend developer	2 days	Location records with timestamps
Develop live map and bus markers	Frontend developer	3 days	Map with last known bus position
Calculate stop ETA and show stale-data state	Full-stack developer	3 days	ETA and last-update indicators
Integration, field and authorization tests	QA + team	3 days	Test results and demo

8. Scrum Events and Definition of Done

Scrum events: Sprint planning establishes the goal and selected backlog; daily Scrum checks progress and blockers; sprint review demonstrates working features to the coordinator and parent representative; retrospective identifies improvements for the next sprint.

Definition of Done: Code reviewed and merged; unit/integration tests pass; acceptance criteria demonstrated; access control verified; errors logged; documentation updated; feature deployed to test environment and approved in review.

9. Risks and Mitigation

Risk	Mitigation
GPS signal or network loss	Show last update time; buffer readings offline and synchronize on reconnection.
Missed student scans	Offer manual confirmation with audit trail and train attendants.
Unauthorized child location access	Use role-based authorization, encryption, short-lived sessions and audit logs.
Overcommitted sprint	Prioritize Must-have stories and revise estimates from observed velocity.

Risk	Mitigation
Unreliable push notifications	Retry failed deliveries and use an approved fallback channel.

10. Expected Outcomes and Conclusion

The Agile plan delivers transport management, live tracking, verified student events and alerts in small testable increments. Sprint reviews ensure that parents and school staff can validate real workflows early. Expected outcomes include improved student safety, better parental visibility, quicker incident response and centralized transport records.

Conclusion: Scrum supports continuous feedback, flexible prioritization and frequent validation of safety-critical functionality for the Smart School Bus Tracking and Student Safety System.